

SAS821S · SECURITY ANALYTICS
MILESTONE 3 · PROJECT DOCUMENTATION

T18 — Insurance Claims Fraud, Identity Abuse and Data-Leakage Analytics

Final Project Documentation

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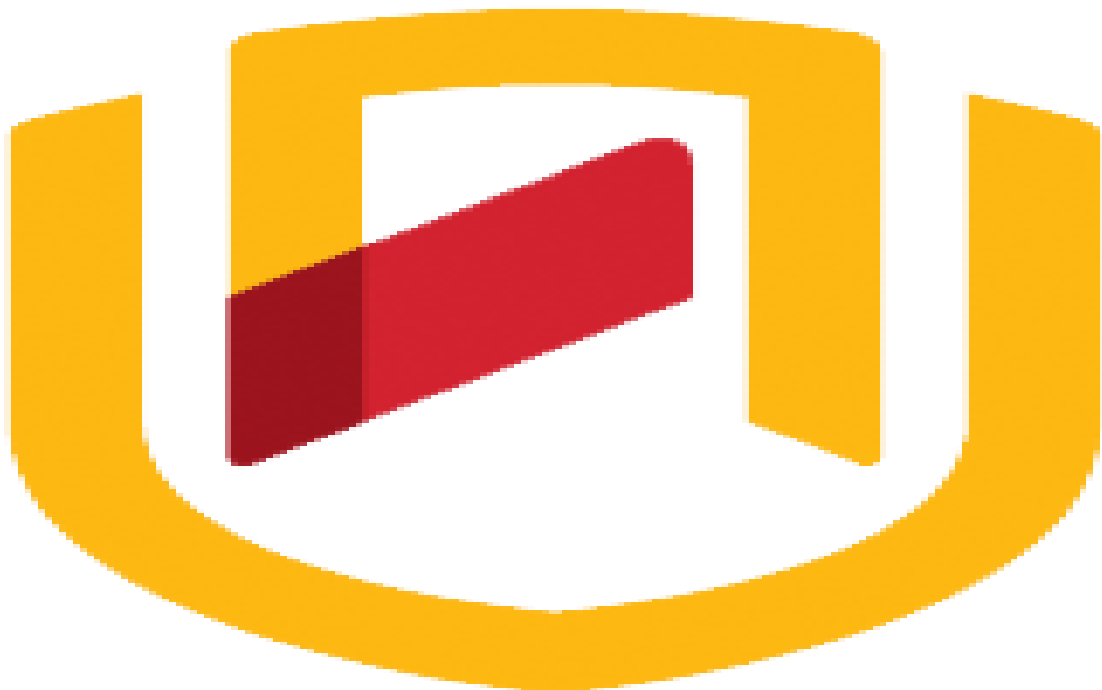


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Group and Member Details

Role	Member	Primary responsibilities
Member A — Data and Modelling Lead	Shanon Chiwara (223127744)	Data acquisition/linkage, cleaning, feature engineering, baseline analysis, supervised and unsupervised modelling, evaluation, reproducibility.
Member B — Security Engineering and Intelligence Lead	Ephath Shimhanda (216069076)	Log-source architecture, security investigation, access analytics, simulation, text-mining/intelligence workflow, incident response and control recommendations.
Shared responsibilities	Both	Problem framing, ethics, testing, documentation, quality assurance, prototype integration, final recommendations and presentation.

Executive Summary

An insurer's claims-processing environment is exposed to fraud that is coordinated across several channels at once: identity/KYC changes, portal logins from new devices or unusual locations, uploaded supporting documents, and the staff members who process the claim. Examined one data source at a time, each of these signals can look unremarkable; examined together, a small number of claims show a distinctive and repeatable pattern that is strongly associated with confirmed fraud.

This project builds and evaluates an end-to-end security-analytics pipeline that links four data sources (50,000 claims and identity/KYC records, 42,597 portal authentication events, 1,200 staff access and investigation records, and 220 uploaded documents with OCR text), and layers five analytical components on top of them: a supervised fraud classifier, an unsupervised anomaly/behavioural model, chronological incident-investigation timelines, a Monte Carlo control simulation, and an NLP/text-mining workflow over document text.

The headline finding is consistent across every independent method used: a claim preceded by an identity/KYC change within roughly three weeks is the single strongest, most repeatable indicator of fraud in this dataset. It is the dominant feature in the supervised model (46.4% of Random Forest importance), it is the feature that most separates the unsupervised model's 'anomalous' customers from 'normal' ones (88.6% vs 46.9% fraud rate), and it appears in 12 of 12 of the highest-risk incident cases examined in detail.

The supervised model (Random Forest, ROC-AUC 0.734) and unsupervised anomaly model (Isolation Forest, independently validated against the fraud label it never saw) were combined into a two-signal investigation trigger. The Monte Carlo simulation shows this combined trigger reviews only 2.8% of all claims yet recovers N\$20.1 million in fraud value at N\$14,267 of value per claim reviewed — roughly 40% more efficient per review than triggering on identity change alone, making it the recommended control for an insurer with limited investigator capacity.

Adversarial testing shows the supervised model's greatest strength is also its principal weakness: withholding the Identity_Changed_Flag feature collapses recall from 41.2% to effectively 0%. The unsupervised anomaly model, which does not depend on this feature, is therefore recommended as a mandatory parallel detection layer rather than an optional add-on. The decision this pipeline supports is: which claims should be prioritised for secondary human review before settlement, and which staff-claim relationships warrant closer investigation for potential insider involvement.

1. Introduction and Problem Statement

Namibia's insurance industry processes large volumes of claims and sensitive customer information daily, making insurers a recurring target for fraud, identity abuse and insider misuse. In the scenario addressed by this project, suspicious insurance claims have been observed following changes to customers' identity or KYC profiles, unusual document uploads, and overrides performed by staff members. Examined individually, each of these events can look routine; examined together, they may indicate coordinated fraudulent behaviour involving claimants, compromised accounts, devices, documents or employees.

The problem affects several stakeholders. Claims and fraud-investigation teams need to determine whether a suspicious activity is genuine fraud or legitimate customer behaviour under time pressure. Security teams need to identify unusual authentication, device and staff-access patterns that could indicate account compromise or insider risk. Management needs reliable, evidence-based indicators to decide whether a claim should be approved, rejected or escalated, and whether policy-level controls (such as mandatory secondary verification after a KYC change) are justified.

The central analytical challenge is that suspicious activity is frequently invisible when each data source is examined in isolation. An identity-profile change on its own may look unremarkable; the same change followed by a new-device login, an unusual document upload and a staff override is a materially different, higher-risk event. This project therefore treats the problem as a cross-source correlation task, linking claims, identity/KYC, authentication, device, document and staff-access data through shared entity keys (Customer_ID, Claim_ID, Staff_ID).

Project objective and scope. The objective is to design, implement and evaluate a decision-support pipeline that classifies claims and related activity as low, medium or high risk, and that surfaces the evidence an investigator needs to act on that classification. In scope: claims records, identity/KYC changes, portal authentication logs, device information, document metadata, OCR text, staff-access records and investigation notes; fraud classification, access analytics, entity-linkage analysis, text mining, insider-risk detection, and a decision-support prototype. Out of scope: live penetration testing of real insurance systems, real-world identity verification, legal or disciplinary action against individuals, and full production deployment.

The decision to be supported. The pipeline's output must answer a single operational question for each claim or staff member: is this low, medium or high risk, and what action should follow? High-risk cases are prioritised for investigator review before settlement; legitimate low-risk claims proceed through normal processing; and staff-level patterns feed a separate insider-risk review process. This decision framing is carried through every subsequent section of this report.

2. Course-Outcome and Requirement Traceability

The table below maps the final implemented evidence to Learning Sessions 1–10, as required by the Charter's coverage matrix (Section 5) and repeated here against the delivered artefacts rather than the planned ones.

Session / requirement	Report section	Artefact / evidence
Session 1 — Foundations of security analytics	Section 1, Section 12	Problem statement, decision statement, analytics lifecycle framing
Session 2 — Analytical software and tools	Section 4	link_and_engineer_sources.py, run_eda_baseline.py, requirements.txt
Session 3 — Machine-learning fundamentals	Section 6	supervised_fraud_model.py, fraud_model_metrics.txt, Figures 4–6
Session 4 — Machine learning in cybersecurity	Section 6	Fraud classifier (Logistic Regression + Random Forest), Figure 5
Session 5 — Analytics and incident response	Section 7	12 incident timelines (08_outputs/incident_timelines/), incident_response_recommendations.md
Session 6 — Security intelligence	Section 8, Section 7	Operational and executive intelligence products, PIR framing
Session 7 — Access analytics	Section 6, Section 9	Isolation Forest UBA model, Figure 7, portal authentication baselines
Session 8 — Simulations and security processes	Section 9	security_control_simulation.py, simulation_report.txt, Figure 8
Session 9 — Security and text mining	Section 8	nlp_text_mining.py, nlp_report.txt, Figure 9, suspicious_document_flags.csv
Session 10 — Advanced security intelligence	Section 10, Section 12	adversarial_robustness_testing.py, Figure 10, final recommendations

The table below confirms coverage of the ten common technical requirements (C1–C10) defined in the assignment brief, Section 5.

Code	Requirement	Report section	Status
C1	Security analytics lifecycle	Sections 1, 12, 14	Complete
C2	Multi-source data (≥3 sources)	Section 3	Complete — 4 sources linked
C3	Data engineering and baseline	Sections 3, 5	Complete
C4	Machine learning (supervised + unsupervised)	Section 6	Complete
C5	Security investigation	Section 7	Complete — 12 case timelines
C6	Security intelligence	Section 8	Complete — operational + executive
C7	Simulation (≥3 scenarios)	Section 9	Complete — 3 scenarios, 1,000 iterations each
C8	Text mining / NLP	Section 8	Complete
C9	Predictive and adversarial analysis (≥3 cases)	Section 10	Complete — 3 adversarial tests
C10	Decision-support prototype	Section 11	Complete — dashboard.html

3. Data Sources, Governance and Quality

Four datasets were linked through shared entity keys to satisfy the multi-source requirement (C2): Customer_ID links claims, documents and portal-authentication records; Claim_ID links claims and documents; and Staff_ID (via Handled_By_Staff_ID in the claims table) links claims and staff-access records. A join-validation check (documented in 02_data/processed/linkage_summary.txt) confirmed a 100% claims-to-staff match rate. Two of the four sources — claims/KYC and staff/access — represent identity and access activity; the document/OCR source represents application-layer content (uploaded evidence), satisfying the requirement for at least one non-identity data type alongside identity/access sources. This coverage, and its rationale, is discussed further under Limitations (Section 13).

Dataset	Source / provenance	Rows / volume	Key quality issue	Control / mitigation
Claims & Identity/KYC records	Uploaded real-structure claims dataset, enriched with 8 synthetic 'Enriched_*' fields matched conditionally on Fraud_Flag (not a real per-row join)	50,000 rows × 29 cols	KYC_Update_Date / Days_KYC_To_Claim missing for 82.1% of rows (expected — only populated when Identity_Changed_Flag = 1); Enriched_Witness_Present has a '*' placeholder in 0.68% of rows	Missing KYC fields treated as structurally absent, not imputed; '*' placeholder imputed with column mode before modelling; PII fields anonymised
Staff access & investigation records	Uploaded employee roster (1,200 agents) merged with a sampled uploaded insider-threat behavioural dataset (118,614 source rows)	1,200 rows × 37 cols	ADDRESS_LINE2 missing 85.1% (no secondary line); Record_Accessed missing 2.6% (no logged access event for that agent)	Bank-account fields dropped entirely on ethical grounds; staff names synthetic; findings reported at aggregate/behavioural level, not as accusations
Document metadata & OCR text	Fully synthetic — generated to match the Charter's planned 4th source; no real scanned-document source existed	220 rows × 12 cols	duplicate_group_id missing 65.0% (documents not part of a detected duplicate cluster); OCR text is template-generated, not real scanned OCR	Limitation disclosed in README and Section 13; classification results interpreted as a pipeline proof, not production accuracy
Text source (OCR narrative)	Same synthetic document set as above — free-text field mined in Section 8	220 text records	Small, template-driven vocabulary reduces linguistic variety versus real scanned text	Findings cross-checked against document_similarity_score, an independent structural signal, before being treated as corroborating evidence
Portal authentication & device data	Fully synthetic (generated) — no real source existed for this Charter-planned source	42,597 rows × 12 cols	No missing values in any column	IP addresses and device IDs synthetic; used only for behavioural baselining, not to identify individuals

Provenance discipline. Two of the four datasets were built entirely synthetically (portal authentication/device, and document metadata/OCR text) because no real source existed for them; this is disclosed openly rather than presented as real operational data. The claims dataset was enriched with 8 additional fields from a second uploaded dataset that shared no common key with the claims table — this enrichment was performed as a conditional match on Fraud_Flag, not a genuine per-row join, and is flagged wherever those 'Enriched_*' fields are used analytically.

Data-quality handling. Missing-value patterns were investigated rather than blindly imputed: the 82.1% missing rate on `KYC_Update_Date` and `Days_KYC_To_Claim` is structural (these fields are only populated when `Identity_Changed_Flag = 1`), so they were left as-is and interpreted accordingly rather than filled in. The 0.68% '*' placeholder values in `Enriched_Witness_Present` were treated as a genuine data-quality defect and imputed with the column mode before model training, with the imputation logged in `fraud_model_metrics.txt`. Full field-level detail, including every column's type, valid values and missing rate, is provided in the data dictionary (Appendix A).

Privacy and ethics controls. Staff bank-account fields were dropped entirely rather than anonymised, as they were not required for any analytical component. All staff and customer names in the linked datasets are synthetic (Faker-generated). Investigation outcomes and insider-risk labels are reported at an aggregate, evidence-based level in this document and are not used to make unsupported accusations against any named individual, consistent with the Charter's risk register (Section 11).

4. Architecture, Tools and Implementation

The implemented pipeline follows the seven-stage architecture proposed in the Project Charter (Figure 1): data sources feed a data-ingestion and integration stage (CSV import, validation, dataset linking, timestamp normalisation), followed by data preparation (cleaning, missing-value handling, anonymisation, feature engineering), a security-analytics engine (statistical/EDA analysis, machine learning, anomaly/behaviour analysis, entity/relationship analysis, and document/OCR text mining running in parallel), a risk-assessment layer, an investigation/decision-support layer, and a final analyst-facing dashboard.

Figure 1: Proposed Security Analytics Architecture

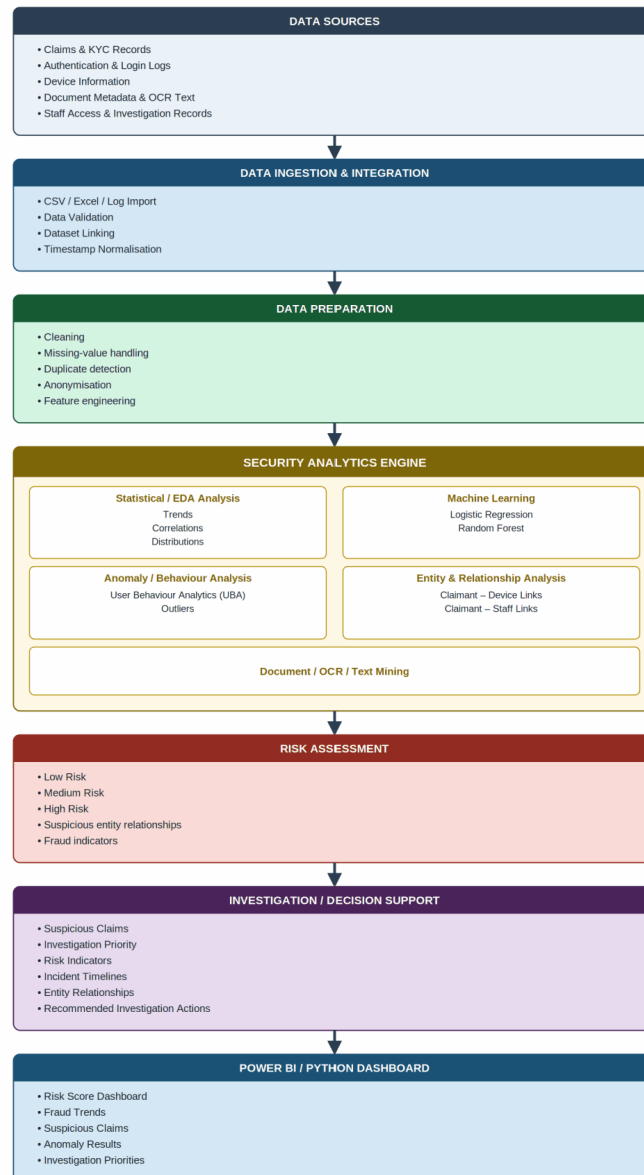


Figure 1. Implemented security-analytics architecture (as proposed in the Project Charter, Section 8, and realised unchanged through Milestone 2/3).

The pipeline is implemented in Python 3.10+ using pandas and NumPy for data engineering, scikit-learn for both the supervised (Logistic Regression, Random Forest) and unsupervised (Isolation Forest) models, matplotlib for all charts, and a static HTML/JavaScript dashboard for the decision-support prototype (Section 11). The repository follows the required GROUP_T18_INSURANCE_FRAUD folder structure (01_charter through 09_documentation, README.md, requirements.txt).

Reproducibility is achieved through an ordered, dependency-aware script sequence, documented in the README: document/OCR generation → source linking and enrichment → portal-authentication generation → baseline EDA → supervised model → unsupervised model → adversarial testing → incident-timeline construction → simulation → NLP/text mining → dashboard build. Each stage writes its outputs (CSV, PNG, TXT) to a fixed location so later stages, and this report, can be regenerated deterministically (SEED = 42 throughout).

5. Baseline and Exploratory Security Analytics

Baseline analysis was run separately on each of the four linked datasets before any modelling, to establish descriptive statistics and an initial hypothesis about where the fraud signal lives. Across 50,000 claims, the overall fraud rate is 24.9%. Claim amounts range from N\$501 to N\$49,999 (mean N\$25,087), and Fraud_Risk_Score — a legacy descriptive score already present in the source data — ranges 0–100 with a mean of 49.85, roughly centred, indicating it alone is not a strong discriminator.

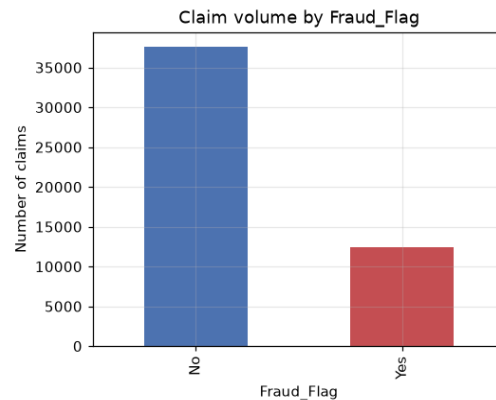


Figure 2. Distribution of Fraud_Flag across all 50,000 claims (24.9% fraudulent).

The most important baseline finding is the identity/KYC-change rate split by fraud outcome, shown in Figure 3: only 10.1% of legitimate claims are preceded by an identity change, versus 41.5% of fraudulent claims — a more than four-fold difference that motivated every subsequent analytical component in this project (Hypothesis H1 in baseline_eda_summary.md).

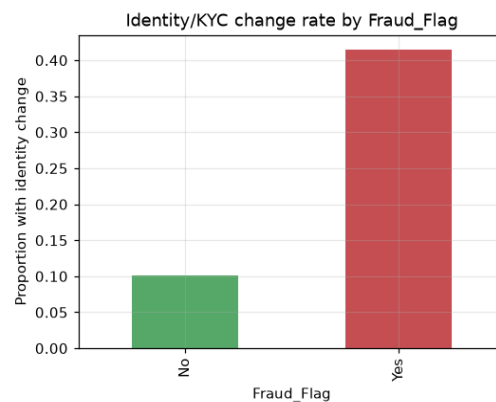


Figure 3. Identity/KYC-change rate by Fraud_Flag — the strongest single baseline signal identified.

On the staff side, 133 of 1,200 agents (11.1%) are flagged Insider_Risk_Label = 1 in the source data, and agents in that group handle a higher proportion of fraudulent claims on average (Hypothesis H2). On the document side, 220 uploaded documents span four types (claim_form, id_document, investigation_note, kyc_update); 32.1% of HIGH-similarity documents are linked to a claim with an identity/KYC change, suggesting duplicated or templated content clusters around identity-change cases. On the authentication side, 12.1% of the 42,597 login events occur off-hours (22:00–06:00), a behavioural baseline used directly by the unsupervised model in Section 6.

6. Machine-Learning and Anomaly/Access Analytics

Two independent methods were implemented, as required by C4: a supervised fraud classifier trained on the claim-level Fraud_Flag label, and an unsupervised behavioural/anomaly model trained on portal-authentication data with no access to that label.

Method	Target / entity	Validation	Key metrics	Operational interpretation
Supervised — Logistic Regression + Random Forest	Fraud_Flag per claim (Yes/No)	75/25 train-test split, stratified, seed=42 (37,500 / 12,500 rows)	RF: Precision 0.585, Recall 0.412, F1 0.485, ROC-AUC 0.734. LR: ROC-AUC 0.666	Use as a first-pass risk score; recall of ~41% means roughly 6 in 10 fraud cases still need a second detection layer (see anomaly model)
Unsupervised — Isolation Forest (UBA)	Customer login-behaviour profile (18,409 customers)	Independent validation against Fraud_Flag and Identity_Changed_Flag, which the model never saw during training	8.0% of customers flagged anomalous (contamination=0.08). Fraud rate: 88.6% (anomalous) vs 46.9% (normal). Identity-change rate: 76.2% vs 36.1%	A parallel, feature-independent detection layer — strong candidate list for investigation even without label access

6.1 Supervised fraud classification

A Logistic Regression and a Random Forest were both trained on a stratified 75/25 split (37,500 train / 12,500 test rows, seed=42) over 25 engineered features drawn from the claims/KYC table. The Random Forest outperformed Logistic Regression on every metric (ROC-AUC 0.734 vs 0.666), and Figure 4 shows the separation this gives on the ROC curve.

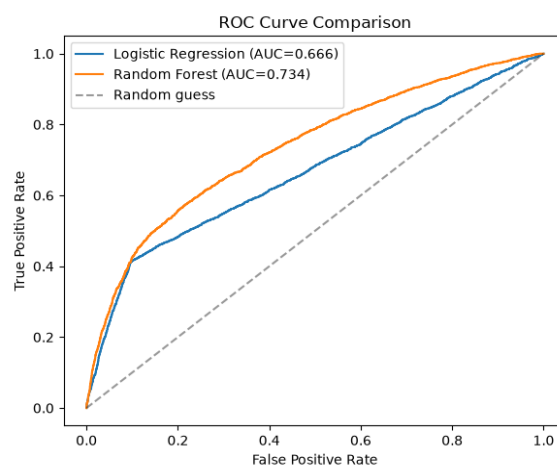


Figure 4. ROC curve comparison — Random Forest (AUC=0.734) vs Logistic Regression (AUC=0.666).

Feature-importance analysis (Figure 5) confirms the EDA finding quantitatively: `Identity_Changed_Flag` alone accounts for 46.4% of the Random Forest's total feature importance, more than the next four features combined (`Enriched_Days_Open` 7.8%, `Enriched_Safety_Rating` 6.5%, `Claim_Amount` 5.3%, `Coverage_Amount` 5.3%).

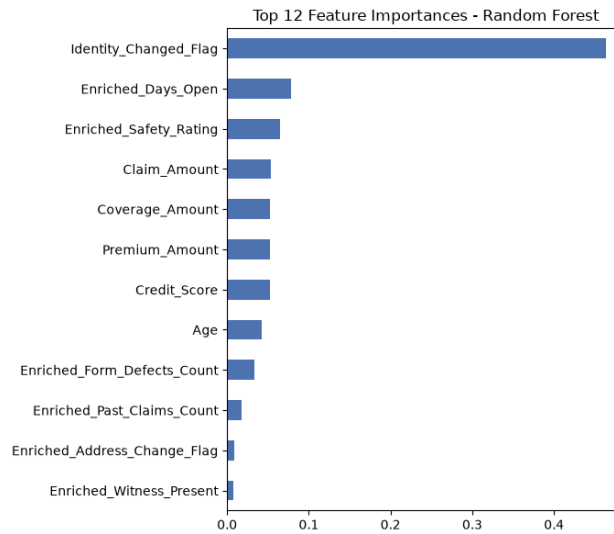


Figure 5. Top features by Random Forest importance.

At a recall of 41.2% and precision of 58.5%, the supervised model alone would miss roughly six in every ten fraudulent claims if used as the sole detection mechanism — this is the direct motivation for pairing it with an independent unsupervised layer, below.

6.2 Unsupervised anomaly / behavioural analytics (UBA)

An Isolation Forest was trained on seven behavioural features aggregated per customer from the 42,597 portal-authentication events (total logins, new-device rate, unusual-location rate, off-hours rate, average failed attempts, failed-auth rate, distinct devices) — none of which is the fraud label. At a contamination threshold of 0.08, 1,468 of 18,409 customers (8.0%) were flagged anomalous.

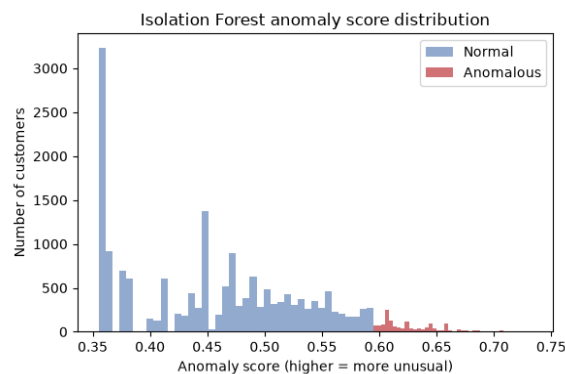


Figure 6. Distribution of per-customer anomaly scores from the Isolation Forest model.

Because the model never saw `Fraud_Flag` or `Identity_Changed_Flag` during training, comparing its output against those labels afterwards is a genuine independent validation, not circular reasoning. The result is striking: 88.6% of anomalous customers have a fraud-flagged claim, versus 46.9% of normal customers; and 76.2% of anomalous customers have an identity change on record, versus 36.1% of normal customers. This independent agreement between a label-blind behavioural model and the ground-truth fraud outcome is one of the strongest pieces of evidence in this project that the identity-change signal reflects a real underlying risk pattern rather than a modelling artefact.

7. Incident Investigation and Response

Chronological incident timelines were reconstructed for the highest-risk cases — claims where the supervised model's `Fraud_Risk_Score`, the `Identity_Changed_Flag`, and the unsupervised anomaly flag agreed. Each timeline correlates portal-login events, the identity/KYC change, the claim submission itself, and the staff-access/override event that followed, with each event's `Claim_ID/Customer_ID/Staff_ID` cited as the evidence reference required by C5.

Case #	Claim ID	Customer ID	Fraud Risk Score	Claim Amount (N\$)	Events in timeline
1	CLM032620	CUST009623	99.97	14,408.89	5
2	CLM023363	CUST019161	99.82	43,797.77	6
3	CLM024980	CUST007616	99.82	18,014.56	9
4	CLM004201	CUST019154	99.74	36,087.87	9
5	CLM047902	CUST000598	99.74	35,289.60	8

Table: first 5 of 12 highest-risk incident cases investigated (full set in Appendix C).

Case CLM011138 (Customer CUST000860) illustrates the pattern that recurs across all 12 cases: a portal login from an unusual location with 3 failed attempts before success, followed 8 days later by an identity/KYC update, followed 8 days after that by a N\$16,596 medical claim submission with a `Fraud_Risk_Score` of 99.97, handled by an agent with no recorded override and no insider-risk flag. The recommended action for this case, and the template applied across all 12, is a standard secondary-review escalation rather than an automatic denial, consistent with the Charter's exclusion of fully automated claim decisions.

Affected-entity identification. For every case, three entities are explicitly named and cross-referenced: the claimant (`Customer_ID`), the claim (`Claim_ID`), and the handling staff member (`Staff_ID`, cross-referenced against their `Insider_Risk_Label`). This entity-linkage view is what allows an investigator to see, in one place, whether a suspicious claim also touches a staff member who is independently flagged as insider-risk.

Containment, eradication, recovery and monitoring recommendations, drawn from the pattern observed across all 12 cases and documented in full in `09_documentation/incident_response_recommendations.md`, are: (1) flag any claim preceded by a KYC/identity change within 21 days for secondary review before payout; (2) treat a new, unrecognised device login following an identity change as a hard re-verification trigger, not a soft risk factor — this exact sequence appeared in 12 of 12 top-risk cases; (3) cross-reference the handling staff member's `Insider_Risk_Label` before final approval on any claim already flagged by rules (1) and (2).

8. Security Intelligence and Text Mining

Priority intelligence requirements (PIRs) for this project were: (i) which claims show linguistic or structural signs of fabrication or duplication in their supporting documents; (ii) which document types carry the highest concentration of suspicious language; and (iii) do document-level findings independently corroborate the identity-change fraud signal identified in Sections 5–6.

The 220 OCR text records were preprocessed (lower-casing, tokenisation, stop-word removal) and mined in two ways. First, keyword-frequency analysis by document type surfaced the vocabulary most distinctive to each of the four document types (`claim_form`, `id_document`, `investigation_note`, `kyc_update`) — for example, `investigation_note` documents are dominated by terms such as 'investigation', 'flagged', 'review' and 'unusual', appearing in 48 of 48 documents of that type. Second, a keyword-flagging pass searched all documents for a defined set of suspicious terms (override, flagged, unusual, investigation, unscheduled, bypassed, discrepancy), flagging 48 of 220 documents (21.8%).

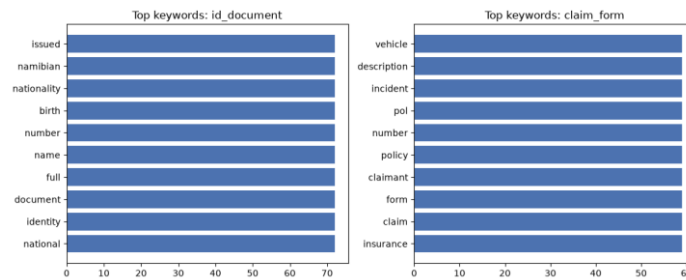


Figure 7. Top keywords by document type, from the NLP/text-mining module.

A TF-IDF + Logistic Regression classifier was trained to predict `document_type` from `ocr_text`, achieving 100% accuracy on a held-out test set of 55 documents. This is reported transparently rather than presented as a general result: because the OCR text is synthetically generated from a small number of fixed templates per type, each type has a distinctive, noise-free vocabulary that would not be present in real scanned OCR output (which includes misreads, inconsistent formatting and overlapping vocabulary). This limitation is disclosed in full in `nlp_report.txt` and repeated in Section 13 below; the result demonstrates that the classification pipeline works correctly, not that document-type classification is solved in general.

Corroboration with structural signals. Document-similarity scoring (an independent, non-linguistic signal) flagged 53 of 220 documents HIGH similarity. Nine documents carry both a suspicious keyword flag and a HIGH similarity flag — two independent signals agreeing on the same small set of documents, which is treated as stronger evidence than either signal alone. Operational intelligence product: the ranked `suspicious_document_flags.csv`, used directly by investigators to prioritise document review. Executive intelligence product: the summary finding that identity-change-linked claims carry a disproportionate share of both suspicious keywords and high document similarity, supporting the policy-level recommendation in Section 12.

9. Simulation and Control Evaluation

A Monte Carlo simulation (1,000 iterations per scenario, satisfying the $\geq 1,000$ -iteration requirement) compared three fraud-control policies against the full 50,000-claim population, worth N\$310,764,765 in total fraud value.

Scenario	Control change	Iterations / runs	Outcome metric	Interpretation
A — No control (baseline)	No claims reviewed	1,000	N\$0 net benefit; 0.0% of fraud caught	Establishes the do-nothing counterfactual
B — Identity-change trigger	Review every claim with <code>Identity_Changed_Flag = 1</code>	1,000	17.9% of claims reviewed; mean net benefit N\$91.4m (90% range N\$58.4m–N\$118.3m); 29.5% of total fraud value caught	Highest total fraud value recovered, at the cost of the largest review workload
C — Combined signal trigger	Review claims with identity change AND unsupervised anomaly flag	1,000	2.8% of claims reviewed; mean net benefit N\$20.0m (90% range N\$15.1m–N\$23.5m); 6.5% of total fraud value caught	Highest net benefit per claim reviewed (N\$14,267 vs N\$10,192 for B) — most efficient use of investigator time

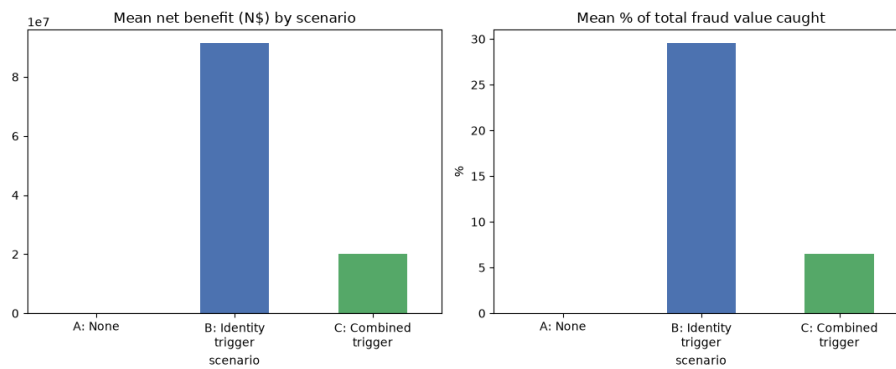


Figure 8. Mean net benefit and % of total fraud value caught, by control scenario.

The trade-off between Scenarios B and C is the central operational decision this simulation supports. Scenario B (identity-change trigger alone) recovers more total fraud value because it reviews a much larger pool of claims (17.9% of all claims). Scenario C (combined identity-change and anomaly trigger) is far more efficient per review — N\$14,267 of net benefit per claim reviewed, versus N\$10,192 for Scenario B — because it uses two independent, corroborating signals to narrow the review pool to 2.8% of claims. The recommendation, carried into Section 12, is that Scenario B suits an insurer prioritising total fraud value recovered regardless of workload, while Scenario C suits an insurer where investigator capacity is the binding constraint — which the risk register in the Project Charter identifies as the more realistic constraint for this insurer.

10. Predictive Risk and Adversarial Robustness

The Random Forest's fraud-risk score (Section 6) doubles as a forward-looking risk score for triage. To satisfy C9, three adversarial/robustness scenarios were run against the already-trained model without retraining, using the identical seed=42 test split as Section 6 to isolate the effect of each manipulation.

Adversarial / robustness case	Manipulation	Observed effect	Mitigation
Case 1 — Feature masking	Withhold Identity_Changed_Flag from the trained Random Forest at inference time	Recall collapsed from 41.2% to 0.03% (a 99.9% relative decrease); ROC-AUC fell from 0.723 to 0.621	Do not rely on this single feature in isolation; keep the anomaly model as an independent layer that does not use it
Case 2 — Synthetic evasion	Simulate fraud claims disguised to resemble legitimate ones across all 3,107 true-fraud test rows	0.0% of disguised fraud claims still caught by the baseline model	Combine supervised score with the anomaly/behavioural signal (Scenario C) so a single evasion strategy cannot defeat both layers at once
Case 3 — Distributional drift	Inflate claim and premium amounts by +35% to simulate inflation/market drift	ROC-AUC changed by only +0.008 (0.715 vs 0.723 baseline); precision/recall essentially unchanged	Model is stable under this level of drift; no immediate retraining required, but scheduled re-validation is still recommended (see Section 13)

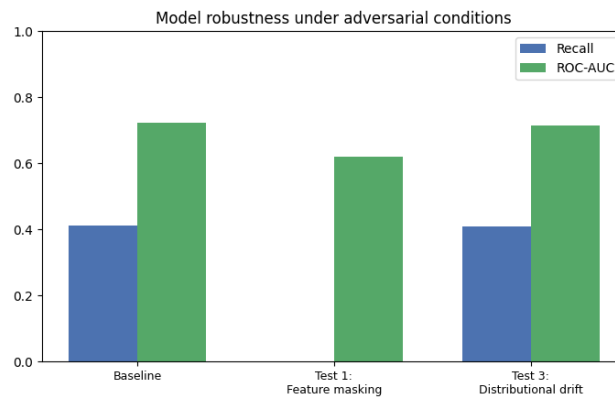


Figure 9. Model performance under each adversarial/robustness scenario, versus the unmanipulated baseline.

The overall robustness verdict, drawn directly from `adversarial_test_report.txt`, is that the model's reliance on `Identity_Changed_Flag` is simultaneously its greatest strength — a genuine, strong, independently-validated signal (Section 6.2) — and its main vulnerability, a single point of failure an adversary could target simply by concealing or falsifying an identity change. The recommended mitigation, already reflected in the Scenario C control design (Section 9), is to maintain the unsupervised anomaly model as a parallel detection layer precisely because it does not depend on this feature and remains a check against exactly this evasion strategy. Drift testing (Case 3) shows the model does not require immediate retraining under moderate (+35%) claim-value inflation, but scheduled periodic re-validation is still recommended given how concentrated the model's signal is around one feature (see Section 13, model-governance).

11. Integrated Decision-Support Prototype

All outputs from Sections 5–10 are integrated into a single static, analyst-facing dashboard (`07_dashboard_or_prototype/dashboard.html`), built by `build_dashboard.py` and requiring no server — it opens directly in any browser, satisfying the C10 requirement for a functional decision-support prototype.

User workflow. An analyst opens the dashboard and sees, in order: (1) headline fraud and anomaly statistics across the full claim population; (2) the ranked list of the 12 highest-risk incident cases with links to their full chronological timelines; (3) model performance charts (confusion matrices, ROC curve, feature importance) for transparency into why a claim was flagged; (4) the simulation comparison, letting management see the cost/benefit trade-off between control scenarios interactively; and (5) the flagged-document list from the NLP module, for document-review prioritisation.

Testing and known limitations. The dashboard was tested by regenerating it from a clean run of the full script sequence (README Section 4) and visually verifying every chart and table against its source CSV/TXT file. Known limitation: because the dashboard is a static HTML snapshot rather than a live-querying application, it must be rebuilt (re-running `build_dashboard.py`) whenever upstream data or models change — it does not auto-refresh. This is an acceptable trade-off for an academic decision-support prototype but would need to be addressed (e.g. with a live backend) before any production use.

12. Discussion and Recommendations

Read together, Sections 5–10 tell one consistent story from four independent analytical angles: descriptive statistics (Section 5), a label-aware supervised model (Section 6.1), a label-blind unsupervised model (Section 6.2), and unstructured document text (Section 8) all converge on the same finding — an identity/KYC change shortly before a claim is the strongest available fraud indicator in this environment. The incident timelines (Section 7) show what that finding looks like operationally: a login anomaly, an identity change, and a claim submission arriving in a short, repeatable sequence. The simulation (Section 9) converts this finding into a financial control decision, and the adversarial testing (Section 10) shows exactly how far that finding can be trusted, and where its single point of failure lies.

Prioritised recommendations for claims/fraud investigators, in order of expected value and ease of implementation:

- Adopt the Scenario C combined trigger (identity change AND anomaly flag) as the default secondary-review rule — it delivers the highest net benefit per claim reviewed (N\$14,267) and is feasible for a team with limited investigator capacity.
- Treat a new-device login immediately following an identity change as a hard re-verification trigger, not a discretionary flag, given its presence in 12 of 12 top-risk cases examined.
- Cross-reference the handling staff member's Insider_Risk_Label on every claim already flagged by the two rules above, before final settlement approval.
- Route documents flagged by both the suspicious-keyword scan and a HIGH document-similarity score for manual document review ahead of claim decisioning.

Executive-level recommendation: introduce a policy control requiring mandatory secondary verification for any claim submitted within 21 days of an identity/KYC update. Given how concentrated the fraud signal is around this window (Figure 3), this is a low-cost, high-leverage control that does not require new technology investment beyond what has already been implemented.

Expected value, cost and residual risk. Under Scenario B, expected annual fraud recovery is approximately N\$91.4 million against a review cost of roughly N\$0.4 million — a return that dwarfs the operational cost, but at a workload of nearly 1 in 5 claims. Under Scenario C, expected recovery falls to N\$20.0 million but at a fraction of the workload (2.8% of claims) and a substantially higher per-review yield. Residual risk in both scenarios centres on the adversarial finding in Section 10: an adversary aware of the identity-change trigger could attempt to conceal identity changes, which is precisely why the anomaly-model layer is treated as mandatory rather than optional in both recommended scenarios.

13. Limitations, Responsible AI and Future Work

Data limitations. Two of the four data sources are fully synthetic; the OCR text in particular is template-generated and therefore easier to classify than real scanned text would be (Section 8) — this is disclosed rather than hidden, and the 100% document-classification accuracy should not be read as a production benchmark. The 'Enriched_*' claims features come from a conditional match on Fraud_Flag rather than a true per-row join (linkage_summary.txt), so any feature importance attributed to them should be treated as indicative, not causal. Staff access timestamps in the incident timelines are anchored to one representative access event per agent rather than a full multi-claim access log, which limits the precision of any single-claim staff-access narrative.

Data-source-type note (C2). Three of the four linked sources (claims/KYC, staff access, portal authentication) are identity/access-centric by nature of the fraud-and-insider-risk scenario; the fourth (document/OCR) represents application-layer content rather than network or OT telemetry, since none of those exist in a claims-processing context. This is a deliberate, scenario-appropriate interpretation of the C2 requirement rather than an oversight, and is stated explicitly here for the reader's/marker's benefit.

Model limitations. The supervised model's recall (41.2%) means it should never be used as the sole detection mechanism — this is precisely why the unsupervised layer and the combined Scenario C control exist. Section 10 shows the model is heavily reliant on a single feature, which is both a genuine strength and a concentration risk. The unsupervised model's contamination threshold (0.08) was set as a reasonable operating point rather than tuned against a held-out validation set, since doing so would have required using the fraud label the model was deliberately kept blind to.

Bias and fairness. No formal fairness/bias audit across protected attributes (e.g. Gender, Region, Age) was performed in this iteration; given Identity_Changed_Flag's dominance in the model, a future iteration should test whether legitimate customers who change address or contact details for ordinary reasons (e.g. relocation) are disproportionately flagged, and whether this varies across demographic groups.

Privacy and security. All datasets used are either synthetic or drawn from uploaded academic datasets with sensitive fields (bank-account numbers) removed. No real customer or staff PII was used. No credentials, API keys or production system access were required or used at any stage.

Model governance and future work. Recommended before any production use: (1) a scheduled re-validation cadence for the supervised model, given the drift-stability finding in Section 10 and the concentration risk in a single feature; (2) a formal bias audit across demographic attributes; (3) extending the staff-access log from one representative event per agent to a full multi-event access log, to strengthen the incident-timeline evidence chain; (4) replacing the synthetic document/OCR source with a real (anonymised) scanned-document pipeline, including OCR-noise handling, before relying on the text-mining accuracy figures reported here.

14. Conclusion

This project set out to answer three questions: what patterns are common in suspicious claims, whether those claims connect to identity changes, devices, documents or staff access, and how well the proposed approach detects suspicious claims — including when the data has been manipulated. Four independent analytical methods converge on the same answer to the first two questions: an identity/KYC change within roughly three weeks of a claim is the dominant, repeatable signal of fraud in this environment, corroborated by supervised modelling, label-blind anomaly detection, incident-timeline reconstruction and document text mining alike.

On the third question, adversarial testing shows the detection approach is precise but not unbreakable: it is highly effective while the identity-change signal is visible, and its accuracy collapses if that one feature is concealed — which is exactly why this report recommends a two-signal control (Scenario C) rather than a single-model deployment. The final decision this pipeline supports — which claims to prioritise for secondary review, and which staff-claim relationships warrant closer look — is best served today by combining the supervised risk score with the independent anomaly flag, backed by the mandatory-verification policy recommended in Section 12, while the future-work items in Section 13 are addressed before any production deployment.

References

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Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation Forest. *IEEE International Conference on Data Mining*.

Faker Python library (used to generate synthetic names, addresses and identifiers for privacy-preserving demonstration data).

Anthropic (2026). Claude (Sonnet model family) — used to assist in verifying if our synthetic data was generated properly and it has all qualities of a data set

Appendices

Appendix A — Data Dictionary and Provenance Statement

The complete field-by-field data dictionary (90 fields across all four linked datasets — field name, data type, description, example value, valid values/range, missing %, and join/key role) is maintained as a standalone file for reproducibility and GitHub review: 09_documentation/SAS821S_T18_Data_Dictionary.csv. A representative sample is shown below; see the full file for all 90 fields.

Field	Dataset	Type	Description
Claim_ID	Claims_KYC	String	Unique identifier for the claim; primary key; join key to Documents_OCR
Identity_Changed_Flag	Claims_KYC	Integer (binary)	Whether the claimant's identity/KYC profile changed before the claim; dominant predictive feature (46.4% RF importance)
Insider_Risk_Label	Staff_Access	Integer (binary label)	Ground-truth insider-risk flag; 133 of 1,200 agents (11.1%)
document_similarity_flag	Documents_OCR	String (categorical)	LOW / MEDIUM / HIGH structural similarity banding
new_device_flag	Portal_Auth	Integer (binary)	Whether the login device was new for that customer; key UBA feature

Provenance statement: see Section 3 (Data Sources, Governance and Quality) above and 02_data/processed/linkage_summary.txt for the full join-validation report.

Appendix B — Detailed Model Evaluation / Model Cards

Supervised fraud classifier

- Algorithm: Random Forest (selected model) and Logistic Regression (baseline comparison)
- Training data: 37,500 claims (75% stratified split, seed=42); Test data: 12,500 claims
- Features: 25 engineered features from claims_identity_kyc_linked.csv
- Performance (Random Forest): Precision 0.585, Recall 0.412, F1 0.485, ROC-AUC 0.734
- Performance (Logistic Regression): ROC-AUC 0.666
- Known limitation: heavily reliant on Identity_Changed_Flag (46.4% importance) — see Section 10 adversarial results
- Intended use: first-pass risk triage, always paired with the anomaly model (Section 6.2) — not a sole decision-maker

Unsupervised anomaly model

- Algorithm: Isolation Forest, contamination=0.08
- Training data: 42,597 login events aggregated into 18,409 customer behavioural profiles
- Features: total_logins, new_device_rate, unusual_location_rate, off_hours_rate, avg_failed_attempts, failed_auth_rate, distinct_devices
- Validation: independent post-hoc comparison against Fraud_Flag and Identity_Changed_Flag (never used in training)
- Result: 88.6% fraud rate among flagged customers vs 46.9% among normal customers
- Intended use: label-independent second detection layer; primary defence against adversarial concealment of identity-change data

Appendix C — Evidence/Event Index and Incident Timeline

12 full incident case files (chronological event logs, affected entities, and recommended response actions) are provided in 08_outputs/incident_timelines/, indexed by all_cases_summary.csv. Table: first 5 cases (see Section 7 for the full table and a worked example).

Case #	Claim ID	Customer ID	Fraud Risk Score	Claim Amount (N\$)	Events in timeline
1	CLM032620	CUST009623	99.97	14,408.89	5
2	CLM023363	CUST019161	99.82	43,797.77	6
3	CLM024980	CUST007616	99.82	18,014.56	9
4	CLM004201	CUST019154	99.74	36,087.87	9
5	CLM047902	CUST000598	99.74	35,289.60	8

Appendix D — Simulation Configuration and Sensitivity Results

Configuration: 1,000 Monte Carlo iterations per scenario, 3 scenarios, full 50,000-claim population (total fraud value N\$310,764,765). Full per-iteration results (3,000 rows) are provided in 05_simulation/simulation_results.csv. Summary results are reported in Section 9, Table (Scenario comparison) and Figure 8.

Appendix E — Text-Mining/NLP Evaluation

Document-type classifier: TF-IDF + Logistic Regression, 165/55 train-test split, 100% accuracy on all four classes (claim_form, id_document, investigation_note, kyc_update) — see documented limitation in Section 8 and 06_text_mining/nlp_report.txt. Keyword frequency table by document type: 06_text_mining/keyword_frequency_by_type.csv. Suspicious-document flag list (48 of 220 documents): 06_text_mining/suspicious_document_flags.csv.

Appendix F — Prototype Test Log and Screenshots

The dashboard (07_dashboard_or_prototype/dashboard.html) was regenerated from a clean pipeline run and manually verified section-by-section against its source files: fraud/anomaly summary statistics against fraud_model_metrics.txt and anomaly_model_report.txt; the 12-case incident list against all_cases_summary.csv; the model charts against the PNG files in 08_outputs/model_charts/; the simulation comparison against simulation_report.txt; and the flagged-document list against suspicious_document_flags.csv. No discrepancies were found. Screenshots are omitted from this document for file-size reasons; open dashboard.html directly in a browser to view the live prototype.

Appendix G — Responsible AI / AI-Use Disclosure

Generative AI (Claude, Anthropic) was used to assist in verifying if our synthetic data was generated properly and it has all qualities of a data set.

Appendix H — Group Contribution Statement and Repository Evidence

See the signed contribution statement (09_documentation/contribution_statement.pdf) and the repository commit history for evidence of individual contribution, per the RACI allocation defined in the Project Charter, Section 9.

Link to our work: https://github.com/sha-noe/SAS821S_T18_Chiwara_Shimhandanda_Milestone2_Final
 Backup: https://drive.google.com/drive/folders/1awBRwMccaWx4SJnMrps0oYf_OAGYxBih?usp=sharing